

# DS358 DSP - Introduction & Sound Units

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# Some Resources for Deep Speech Processing

- L.R. Rabiner and R.W. Schafer, Digital Processing of Speech Signals Pearson Education, Delhi, India, 2004
- L. R. Rabiner, B. H. Juang and B. Yegnanarayana, "Fundamentals of speech recognition", Pearson Education, 2009.
- Linus Ericsson, et al., Self-Supervised Representation Learning: Introduction, Advances and Challenges, arXiv:2110.09327
- R. Prabhavalkar, et.al., "End-to-End Speech Recognition: A Survey," in IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 32, pp. 325-351, 2024,
- Xie, Tianxin, et al. "Towards Controllable Speech Synthesis in the Era of Large Language Models: A Survey." (2024).
- Wang, S., et.al., "Overview of speaker modeling and its applications: From the lens of deep speaker representation learning" IEEE/ACM Transactions on Audio, Speech, and Language Processing.



- **Overview of Speech Processing:** Speech production, perception and comprehension, spoken language processing, human speech processing tasks, nature of speech signal, signal processing of speech, linguistics and phonetics, feature extraction. (6 hrs)
- **Machine Learning of Speech Processing:** clustering of speech features, classification tasks in speech, Gaussian mixture model (GMM), Hidden Markov Model (HMM), Support Vector Machine (SVM) and Neural Networks (NN) (6 hrs)
- **Deep Learning of Speech Processing:** Self-Supervised Learning for Speech Representation. Automatic Speech Recognition (ASR). Text to Speech (TTS) synthesis. Speaker and Language Recognition. (Details in Annexure - DSP) (30 hrs)

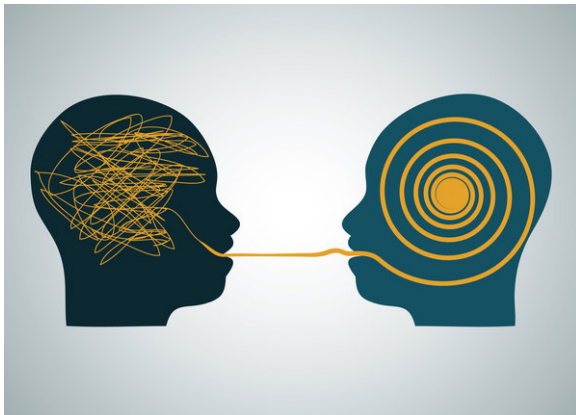
# Organization

- Human speech processing
- Natural Language : Spoken vs written
- Information sources in speech
- Human speech processing tasks
- Acoustic form of speech signal
- Electrical form of speech signal

# Human Speech Processing

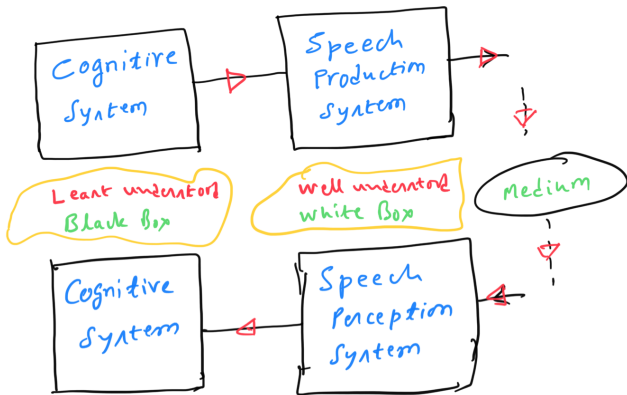
- Cognitive process
- Physiological process
- Production of speech
- Perception of speech
- Comprehension of speech

# Cognitive Processing



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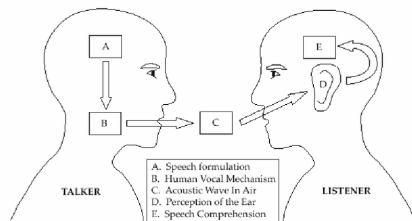
# Cognitive and Physiological Processes



# Human - Human Communication

|                 | <b>Verbal Communication</b>        | <b>Nonverbal Communication</b>     |
|-----------------|------------------------------------|------------------------------------|
| <b>Oral</b>     | Spoken Language                    | Laughing, Crying, Coughing, Etc... |
| <b>Non Oral</b> | Written Language/<br>Sign Language | Gestures, Body Language, Etc...    |

(a) Verbal vs Non-Verbal Communication<sup>1</sup>

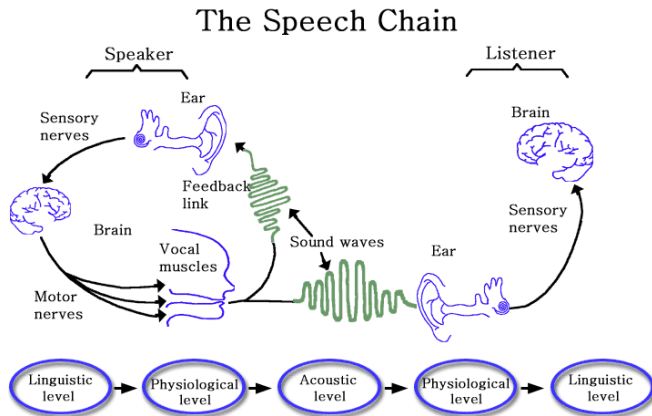


(b) Speech production, transmission, perception, comprehension<sup>2</sup>

1.[<https://mytext.cnm.edu/lesson/5-1-0-defining-verbal-communication/>]

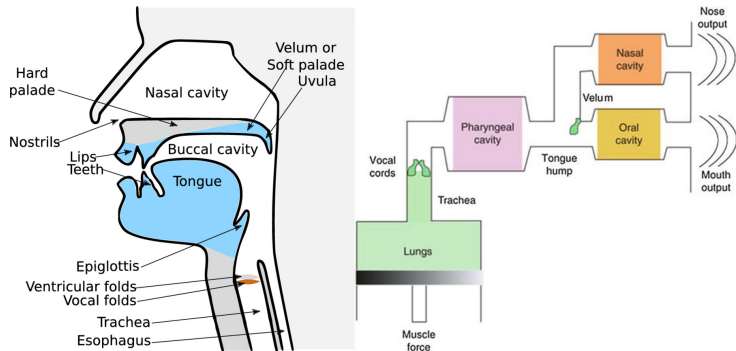
2.[<https://towardsdatascience.com/speech-recognition-is-hard-part-1-258e813b6eb7>]

# Human Speech Communication Chain



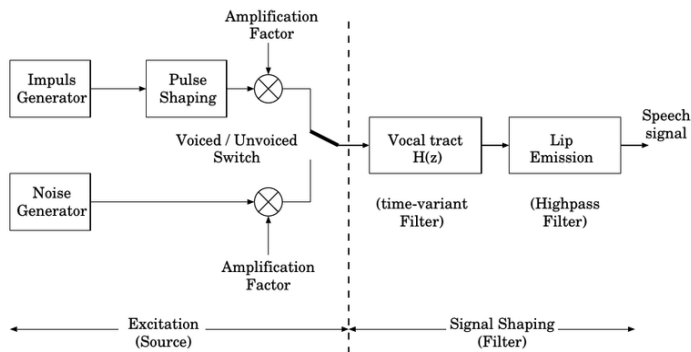
<http://indra-bohara.blogspot.com/2010/10/brief-critical-review-of-speech-chain.html>

# Feature Extraction for Acoustic Modeling



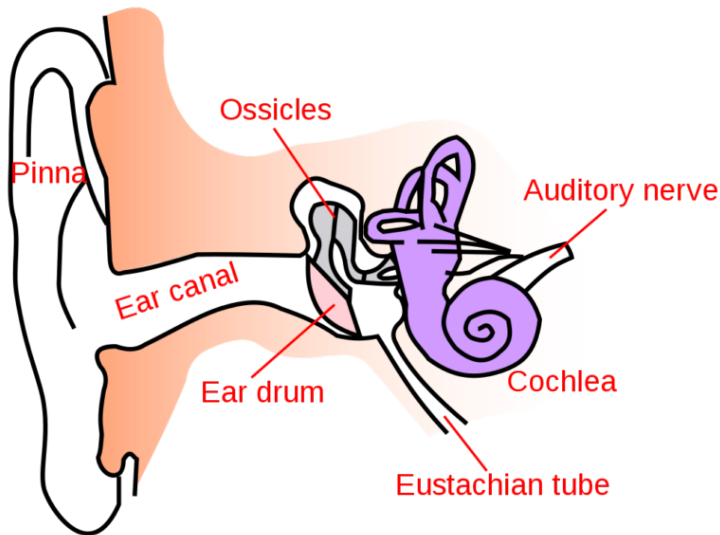
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# Feature Extraction for Acoustic Modeling



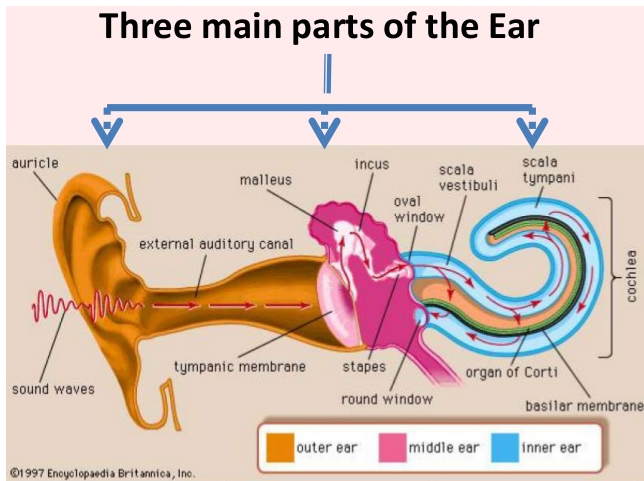
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# Speech Perception Process: Ear



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# Speech Perception Process: Acoustic to BEP



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# Natural Language

- Unless specified language refers to **natural language** used by humans
- Not to be confused with **artificial languages** used by computers
- A natural language has developed and evolved naturally, through use by human beings
- As opposed to an invented or constructed language, as a computer programming language

# Classification of Natural Language

- **Language** : Fundamental method of human communication, consisting of words used in a structured, and conventional way and conveyed by speech, writing, or gesture.
- **Spoken Language** : Human communication via speech mode
- **Written Language** : Human communication via written mode
- **Sign Language** : Human communication via gesture mode

# Components of Natural Language

- **Components of a language** : Basic units for representation, words, rules for forming phrases and sentences
- Meaning of words
- Meaning of phrases
- Meaning of sentences
- Hidden information like emotions, sentiments ...

# Components of Spoken Language

- **Phonemes or phones** : Basic units of speech
- **Syllable** : Bigger basic units of speech, combination of one or more phonemes
- Spoken words, phrases and sentences
- **Prosody** : Elements of speech that are properties of syllables and larger units of speech, including linguistic functions such as intonation, stress, and rhythm

# Hindi Sound Units

- Vowels & consonants
- Short vowels, long vowels and diphthongs
- Stop consonants, nasals, semivowels, fricatives, affricates
- Compound characters

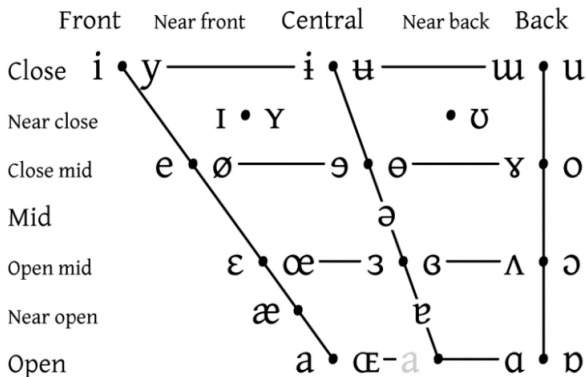
# Hindi Vowels



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# Production of Vowels

## VOWELS



Vowels at right & left of bullets are rounded & unrounded.

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# Hindi Consonants

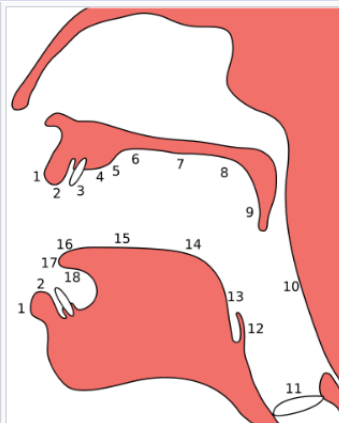


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# Different Aspects during Production

- Place of articulation (PoA)
- Manner of articulation (MoA)
- Type of vowel
- PoA : Glottal, velar, alveolar, palatal, dental, labial
- MoA : Unvoiced unaspirated (UVUA), Unvoiced aspirated (UVA), Voiced unaspirated (VUA) and Voiced aspirated (VA)
- Vowel : Front, back and middle

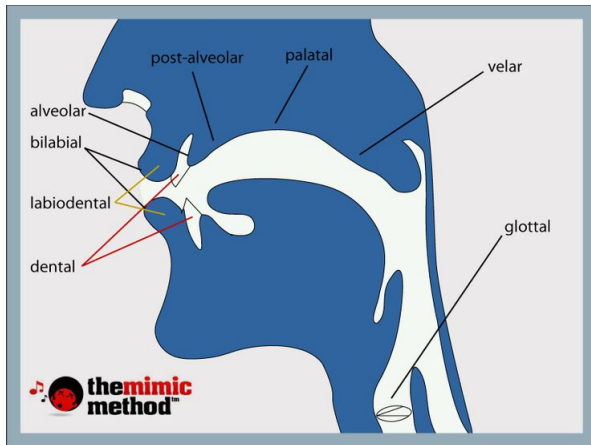
# Place of Articulation



Places of articulation (passive & active): 

1. Exo-labial, 2. Endo-labial, 3. Dental, 4. Alveolar, 5. Post-alveolar, 6. Pre-palatal, 7. Palatal, 8. Velar, 9. Uvular, 10. Pharyngeal, 11. Glottal, 12. Epiglottal, 13. Radical, 14. Postero-dorsal, 15. Antero-dorsal, 16. Laminal, 17. Apical, 18. Sub-apical

# Place of Articulation



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# Hindi Stops - PoA & MoA

|           | Voiceless Unaspirated |                | Voiceless Aspirated |                   | Voiced Unaspirated |             | Voiced Aspirated |               | Nasal |           |
|-----------|-----------------------|----------------|---------------------|-------------------|--------------------|-------------|------------------|---------------|-------|-----------|
| Velar     | क                     | ka<br>/k/      | ख                   | kha<br>/kʰ/       | ग                  | ga<br>/g/   | घ                | gha<br>/gʱ/   | ङ     | ṅa<br>/ŋ/ |
| Palatal   | च                     | ca<br>/c, t͡ʃ/ | छ                   | cha<br>/cʰ, t͡ʃʰ/ | ज                  | ja<br>/d͡ʒ/ | झ                | jha<br>/d͡ʒʱ/ | ञ     | ña<br>/ɲ/ |
| Retroflex | ट                     | ṭa<br>/ʈ/      | ठ                   | ṭha<br>/ʈʰ/       | ड                  | ḍa<br>/ɖ/   | ढ                | ḍha<br>/ɖʱ/   | ण     | ṇa<br>/ɳ/ |
| Dental    | त                     | ta<br>/t/      | थ                   | tha<br>/tʰ/       | द                  | da<br>/d̪/  | ध                | dha<br>/d̪ʱ/  | न     | na<br>/n/ |
| Bilabial  | प                     | pa<br>/p/      | फ                   | pha<br>/pʰ/       | ब                  | ba<br>/b/   | भ                | bha<br>/bʱ/   | म     | ma<br>/m/ |

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# Hindi Semivowels

य र ल व

Y

R

L

V

SEMI VOWELS

| Semivowels | Hindi | Punjabi | Bengali | Gujarati |
|------------|-------|---------|---------|----------|
| Y          | य     | ਯ       | য়      | ય        |
| R          | र     | ਰ       | র       | ર        |
| L          | ल     | ਲ       | ল       | લ        |
| V          | व     | ਵ       | —       | વ        |
| Z          | ज़    | ਜ       | য       | —        |

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# Hindi Fricatives

श

*śa*

ष

*ṣa*

स

*sa*

ह

*ha*

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# Hindi Half-Consonants

## Half consonants – Halant (हलन्त)

|    |    |    |    |    |
|----|----|----|----|----|
| क् | ख् | ग् | घ् | ङ् |
| च् | छ् | ज् | झ् | ञ् |
| ट् | ठ् | ड् | ढ् | ण् |
| त् | थ् | द् | ध् | न् |
| प् | फ् | ब् | भ् | म् |
| य् | र् | ल् | व् |    |
| श् | ष् | स् | ह् |    |

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# Hindi Vowels & Consonants

| Vowels   |     | Consonants |    |   |     |     |
|----------|-----|------------|----|---|-----|-----|
| अ        | a   | क          | ख  | ग | घ   | ङ   |
| आ ( ा )  | A   | k          | kh | g | gh  | n   |
| इ ( ि )  | i   | च          | छ  | ज | झ   | ञ   |
| ई ( ि )  | I   | c          | ch | j | jh  | n   |
| उ ( उ )  | u   | ट          | ठ  | ड | ढ   | ण   |
| ऊ ( ू )  | U   | T          | Th | D | Dh  | N   |
| ऋ ( ृ )  | R   | त          | थ  | द | ध   | न   |
| ॠ ( ॡ )  | RR  | t          | th | d | dh  | n   |
| लृ ( ॢ ) | lr. | प          | फ  | ब | भ   | म   |
| ए ( े )  | e   | p          | ph | b | bh  | m   |
| ऐ ( ै )  | ai  | य          | र  | ल | व   |     |
| ओ ( ो )  | o   | y          | r  | l | v/w |     |
| औ ( ौ )  | au  | श          | ष  | स | ह   | ज्ञ |
| अं ( ं ) | m.  | S          | sh | s | h   | jn  |
| अः ( ः ) | :   |            |    |   |     |     |

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# Summary

- Human speech processing
- Speech as cognitive processing
- Speech as physiological processing
- Speech production, perception and comprehension
- Natural Language : Spoken vs written
- Information sources in speech
- Sounds units
- Classification of sound units

*Thank You*